FEDERAL SPECIFICATION

GRID, RADIOGRAPHIC, STRAIGHT, WAFER-TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** — This specification covers straight wafer-type radiographic grids, complete with carrying case, for use with X-ray apparatus during radiographic examination.

1.2 **Classification.**

1.2.1 *Sizes.*—Grids shall be of the following sizes, as specified (see 6.1).

Size 1.—10 by 12 inches.
Size 2.—14 by 17 inches.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

C–F–206—Felt, Sheet, Wool, Pressed.
NN–P–530—Plywood; Flat-Panel.
TT–E–527—Enamel, Synthetic, Lustreless.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Boxes, Fiber.
PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 *Grids.*—Grids shall be constructed of materials hereinafter specified (see 3.2.1), of the highest quality throughout, and suitable in all respects for the purpose intended.

3.1.2 *Carrying case.*

3.1.2.1 Plywood shall be type I, Douglas fir, conforming to Federal Specification NN–P–530.

3.1.2.2 *Felt.*—Felt shall be an oil-free wool felt conforming to Federal Specification C–F–206.

3.1.2.3 *Other materials.* — Steel shall be used for the channel. The hinge and other hardware shall be fabricated of materials suitable in all respects for the purpose intended.

3.2 Grid.

3.2.1 *Design.*—Grids shall have uniformly spaced strips with angulation for nominal focus at 38 inches, so as to provide for a working range of 34 to 44 inches with minimum transmission through the outer edges of the grid to be not less than 70 percent of the transmission through the center of the grid. The width of the lead strips shall be no less than 0.075 inch and the spacing between them shall be no greater than 0.015 inch, the grid ratio being 6 to 1. The lead thickness shall be at least 0.0017 and shall be no greater than 0.003 inch. There shall be no less than 55 lead strips per inch. The interspace material shall be nonmetallic to provide minimal absorption of the X-ray beam. The lead strips and their dividers shall be firmly cemented between two sheets of aluminum, each 0.013-inch maximum thickness. The aluminum covers shall be roll-crimped around the entire periphery and shall have corners with a maximum radius of ½ inch.

3.2.2 *Performance.*—A radiographic test in accordance with 4.4.1 and identified by the grid serial number shall be submitted with each grid. Marked variations in density, opaque inclusions, lead turnover, doubling or omission of the lead strip, or any other defect which may cause difficulty in radiographic diagnosis shall be considered cause for rejection. Grids showing sectional pattern will not be acceptable.

3.3 Carrying case.—Unless otherwise specified, a carrying case which shall meet the requirements of the drop test specified in 4.4.2, shall be provided for the grid. The carrying case may be fabricated of metal or plywood. If fabricated of plywood, the case sides shall be constructed of ⅜-inch plywood, as specified in 3.1.2.1, sound on one side. The plywood case shall be bound around the narrow sides with a steel channel. The inside of the case shall be padded and completely lined with felt, as specified in 3.1.2.2, to protect the grid from damage. The case shall open on one end and shall be so designed that, when closed, the grid is held firmly to prevent movement. The hinge and the hardware shall be designed and so placed that there will be no projections on the large sides which will interfere with stacking. As an alternate, the carrying case may be designed in such a manner that the case opens bookwise on the large side.

3.4 Finish.—The carrying case hardware shall be pretreated with a phosphate coating and then shall be dipped or sprayed all over with a synthetic primer. A suitable primer shall be applied to the outside of the carrying case prior to application of the paint. The outside of the case, including hardware shall be finished with two coats of enamel conforming to Federal Specification TT–E–527. The color shall be 34087, lusterless, olive drab, as shown in Federal Standard No. 595 (see 6.3).

3.5 Identification marking.

3.5.1 *Grid.*—Each grid shall be marked in a conspicuous location with the stock number,

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the manufacturer's name or registered trademark, serial number, and contract or order number.

3.5.2 *Carrying case.*—Each carrying case shall be permanently marked with the stock number, item identification, and the letters "U.S." in suitable size on one large side and on one end of the carrying case.

3.6 **Workmanship.**—Workmanship shall be first class throughout. The grid and carrying case shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The provisions contained in this section are applicable to inspection performed by and for the Government. Sampling, inspection, and test procedures for components or materials described shall be in accordance with this specification and with specifications and standards referenced herein to the extent applicable, except that this specification shall govern in the event of conflict.

4.1.1 *Supplier responsibilities for inspection for military procurement.*—Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.2 Inspection, as used in this specification, is defined as both examination (such as, visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 **Certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and the test results obtained and indicating that material is as specified, may be furnished.**

4.2 **Sampling.** — **Sampling shall be conducted in accordance with Military Standard MIL–STD–105 and Appendix thereto.**

	Inspection level	AQL (percent defective)
For visual examination:		
Major A	II	1.0
Total	II	4.0
Major B	II	1.0
Total	II	4.0

	Inspection level	Acceptance number	Sample size
For end item testing:			
Radiographic test	---	0	All units procured
Drop test	L5	0	Not less than 3, nor more than 10

4.3 **Inspection.**—The grids and carrying case shall be inspected to determine compliance with all requirements contained in sections 3 and 5 of this specification. Nonconformance with these requirements will be permitted to the extent indicated in 4.2.

4.3.1 *Classification of defects.*—Inspection shall be conducted in accordance with the following classification of defects:

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*Categories and defects**

Major A: (*Unit of product—grid*)

101. Aluminum covers not free of cracks, tears, and deformed surfaces.
102. Lead strips and their dividers not securely cemented between sheets of aluminum.
103. Grid not flat on top and bottom surfaces.

Major B: (*Unit of product—carrying case*)

151. Carrying case fails to open and close properly.
152. Grid fails to fit carrying case.
153. Projections on carrying case which interfere with stacking.
154. Steel channel not bound around sides of plywood carrying case.

Minor A: (*Unit of product—grid*)

201. Dimensions of grid not as specified.
202. Grid not free of sharp edges.

Minor B: (*Unit of product—carrying case*)

251. Surface of carrying case not free of splints, sharp edges, or scratched areas.
252. Identification marking not present, legible, and permanent (in whole, or in part).
253. Labels not legible and free from defacing marks, scratches, and tears.

* Inspection is not restricted to the classified possible defects listed above.

4.4 **Tests.**

4.4.1 *Radiographic test.*—The radiograph shall be made at a target-to-film distance of 38 inches with central X-ray beam vertical to the plane of the grid and centered on the grid.

4.4.2 *Drop test.*—The carrying case with grid in place shall be capable of withstanding the following tests without damage to the case or grid:

(*a*) Four 18-inch drops to a concrete floor, once on each short edge of the case, with the center of gravity directly over the edge on impact.

(*b*) Two 36-inch drops to a concrete floor, with the case landing flat, once on each large side.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing used are defined in Federal Standard No. 102.

5.1 **Packaging.**

5.1.1 *Level A.*

5.1.1.1 *Unit of issue.*—One grid, of the stock number specified, constitutes one unit.

5.1.1.2 *Unit package.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636, type I, class 1. Each carton shall be adequately secured to prevent accidental opening.

5.1.2 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 **Packing.**

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Each unit carton shall be overpacked in an exterior container constructed in accordance with Federal Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Each unit carton shall be overpacked in an exterior container constructed in accordance with Federal Specification PPP–B–621, table II.

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5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind, commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or order.

5.3 Marking.

5.3.1 *Military agencies.* — Exterior container shall be marked in accordance with Military Standard MIL-STD-129, except that 6.2(*c*), 6.2(*d*), and 6.5 shall not be applicable. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

5.3.2 *Civil agencies.*—In addition to any special marking in the contract or order exterior shipping containers shall be marked in accordance with Federal Standard No. 123.

6. NOTES

6.1 Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.
(*b*) Size required (see 1.2.1).
(*c*) If carrying case is not required (see 3.3).
(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items appearing in the Armed Services Medical Stock List:

6525-603-1250 GRID, RADIOGRAPHIC, 10 by 12 inches, Straight Wafer Type.
6525-603-1310 GRID, RADIOGRAPHIC, 14 by 17 inches, Straight Wafer Type.

6.3 **Availability of color chips.**—The individual 3- by 5-inch chip of the color referenced in 3.3 is available from General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. Price 5 cents.

6.4 **Transportation description.**—Transportation description and minimum weight applicable to this commodity are:

Rail:

Photographers' Materials, not otherwise indexed by name.

Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Photographers' materials, not otherwise indexed:

Motor volume minimum weight 20,000 pounds.

6.5 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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